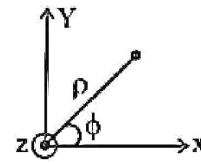


Point Conversion

1. Cartesian to cylindrical

$$\rho = \sqrt{X^2 + Y^2}, \quad \phi = \tan^{-1}\left(\frac{Y}{X}\right), \quad Z = Z$$



2. Cylindrical to Cartesian

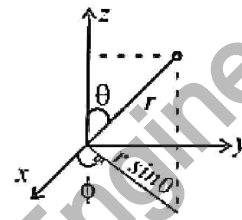
$$X = \rho \cos \phi, \quad Y = \rho \sin \phi, \quad Z = Z$$

3. Cartesian to spherical

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \cos^{-1}\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right)$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$



4. Spherical to Cartesian

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

5. Cylindrical to Spherical

$$r = \sqrt{\rho^2 + z^2}, \quad \theta = \cos^{-1}\left(\frac{z}{\sqrt{\rho^2 + z^2}}\right), \quad \phi = \phi$$

6. Spherical to Cylindrical

$$\rho = r \sin \theta, \quad \phi = \phi, \quad z = r \cos \theta$$

Unit vector Conversion

1. Cartesian to Cylindrical

$$\begin{bmatrix} \hat{a}_\rho \\ \hat{a}_\phi \\ \hat{a}_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{a}_x \\ \hat{a}_y \\ \hat{a}_z \end{bmatrix}$$

2. Cylindrical to Cartesian

$$\begin{bmatrix} \hat{a}_x \\ \hat{a}_y \\ \hat{a}_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{a}_\rho \\ \hat{a}_\phi \\ \hat{a}_z \end{bmatrix}$$

3. Cartesian to Spherical

$$\begin{bmatrix} \hat{a}_r \\ \hat{a}_\theta \\ \hat{a}_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} \hat{a}_x \\ \hat{a}_y \\ \hat{a}_z \end{bmatrix}$$

4. Spherical to Cartesian

$$\begin{bmatrix} \hat{a}_x \\ \hat{a}_y \\ \hat{a}_z \end{bmatrix} = \begin{bmatrix} \sin\theta\cos\phi & \cos\theta\cos\phi & -\sin\phi \\ \sin\theta\sin\phi & \cos\theta\sin\phi & \cos\phi \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} \hat{a}_r \\ \hat{a}_\theta \\ \hat{a}_\phi \end{bmatrix}$$

5. Spherical to Cylindrical.

$$\begin{bmatrix} \hat{a}_\rho \\ \hat{a}_\phi \\ \hat{a}_z \end{bmatrix} = \begin{bmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\sin\theta & 0 \end{bmatrix} \begin{bmatrix} \hat{a}_r \\ \hat{a}_\theta \\ \hat{a}_\phi \end{bmatrix}$$

6. Cylindrical to Spherical

$$\begin{bmatrix} \hat{a}_r \\ \hat{a}_\theta \\ \hat{a}_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta & 0 & \cos\theta \\ \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \hat{a}_\rho \\ \hat{a}_\phi \\ \hat{a}_z \end{bmatrix}$$

1.3. Differential Length, Area and Volume

Parameter			Coefficient		
u	v	w	h_1	h_2	h_3
x	y	z	1	1	1
ρ	ϕ	z	1	ρ	1
r	θ	ϕ	1	r	$r\sin\theta$

1. Differential Length

It is a vector quantity and directed along tangential direction.

In general form,

$$\overline{dl} = h_1 du \hat{a}_u + h_2 dv \hat{a}_v + h_3 dw \hat{a}_w$$

(a) Cartesian co-ordinate system.

$$\overline{dl} = dx \hat{a}_x + dy \hat{a}_y + dz \hat{a}_z$$

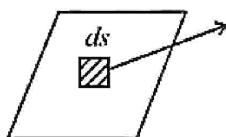
(b) Cylindrical Co-ordinate system.

$$\overline{dl} = d\rho \hat{a}_\rho + \rho d\phi \hat{a}_\phi + dz \hat{a}_z$$

(c) Spherical co-ordinate system.

$$\overline{dl} = dr \hat{a}_r + r d\theta \hat{a}_\theta + r \sin\theta d\phi \hat{a}_\phi$$

2. Differential Area



$$d\vec{s} = ds \hat{a}_n$$

$\hat{a}_n$ = unit normal to the surface

It is a vector quantity and directed normal to the surface.

In general Form

$$d\vec{S} = h_2 h_3 dv dw \hat{a}_u + h_1 h_3 du dw \hat{a}_v + h_1 h_2 du dv \hat{a}_w$$

(a) Cartesian co-ordinate system.

$$d\vec{S} = dydz\hat{a}_x + dx dz\hat{a}_y + dx dy\hat{a}_z$$

(b) Cylindrical co-ordinate system.

$$d\vec{S} = \rho d\phi dz \hat{a}_\rho + \rho dz d\phi \hat{a}_\phi + \rho d\rho d\phi \hat{a}_z$$

(c) Spherical co-ordinate system.

$$d\vec{S} = r^2 \sin \theta d\theta d\phi \hat{a}_r + r \sin \theta dr d\phi \hat{a}_\theta + r dr d\theta \hat{a}_\phi$$

3. Differential Volume

In general form

$$dv = h_1 h_2 h_3 du dv dw$$

(a) Cartesian co-ordinate system :-

$$dv = dx dy dz$$

(b) Cylindrical co-ordinate system :-

$$dv = \rho d\rho d\phi dz$$

(c) Spherical co-ordinate system.

$$dv = r^2 \sin \theta dr d\theta d\phi$$

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C	Formulae
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Mathematical functions, tables and transforms list.

I. Some Mathematical Functions :

Unit Step	$u(t) = \begin{cases} 1 & \text{for } t > 0, \\ 0 & \text{for } t < 0. \end{cases}$
Sin C	$S_b(x) = \frac{\sin x}{x}$
Rectangular	$\text{Rect}\left(\frac{t}{T}\right) = \begin{cases} 1 & \text{for } t \leq T/2 \\ 0 & \text{for } t > T/2 \end{cases}$
Triangular	$\wedge\left(\frac{t}{T}\right) = \begin{cases} 1 - \frac{ t }{T} & \text{for } t \leq T \\ 0 & \text{for } t > T \end{cases}$
Signum	$\text{sign}(t) = \begin{cases} 1 & \text{for } t > 0 \\ -1 & \text{for } t < 0 \end{cases}$
Impulse	$\delta(t) = \begin{cases} 1 & \text{for } t = 0 \\ 0 & \text{for } t \neq 0 \end{cases}$
Bessel function	$I_n(\beta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(\beta \sin \theta - n\theta)} d\theta$
Variance of X	$\sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 P_X(x) dx ;$ in which $\mu = E[X]$, $E \rightarrow$ Expectation

II. Some Probability Functions :**a) Discrete distribution :**

Binomial :

$$P_r(m) = \binom{n}{m} p^m \cdot q^{n-m}; \quad m = 0, 1, 2, \dots, n.$$

$$= 0 \quad \text{otherwise.}$$

$$0 < p < 1, \quad q = 1 - p$$

$$p(x) = \sum_{m=0}^n p^m q^{n-m} \delta(x-m)$$

$$\bar{x} = np \text{ and}$$

$$\sigma_x^2 = npq.$$

Poisson :

$$P_r(m) = \frac{\lambda^m \cdot e^{-\lambda}}{m!}; \quad m = 0, 1, 2, 3, \dots$$

$$p(x) = \sum_{m=0}^n \frac{\lambda^m \cdot e^{-\lambda}}{m!} \delta(x-m)$$

$$\bar{x} = \lambda \text{ and}$$

$$\sigma_x^2 = \lambda$$

b) Continuous distribution :

Gaussian (Normal)

$$p(x) = \frac{1}{\sigma_x \sqrt{2\pi}} \cdot \exp\left[-\frac{(x-\bar{x})^2}{2\sigma_x^2}\right]; \quad -\infty \leq x \leq \infty$$

$$E\{(x-\bar{x})^2\} = \sigma_x^2 \text{ and } E\{x\} = \bar{x}$$

Exponential

$$p(x) = be^{-bx} \quad x > 0$$

$$= 0 \quad \text{otherwise.}$$

and $\sigma_x^2 = a^{-2} \text{ and } \bar{x} = a^{-1}$

c) Rayleigh distribution :

The probability density function of the envelope of the Gaussian Random noise signal with variance σ_n^2 and zero mean.

$$P(a) = \frac{a}{\sigma_n^2} \exp\left(-\frac{a^2}{2\sigma_n^2}\right); \quad a \geq 0$$

$$E\{a\} = \bar{a} = \sigma_n \sqrt{\pi/2}$$

$$E\{(a-\bar{a})^2\} = \sigma_a^2 = \left(2 - \frac{\pi}{2}\right) \sigma_n^2$$

d) Uniform distribution

$$P(a) = \begin{cases} \frac{1}{y-x} & x < a < y \\ 0 & \text{elsewhere} \end{cases}$$

$$\sigma_x^2 = \frac{(y-x)^2}{12} \quad \text{and}$$

$$\bar{x} = \frac{x+y}{2}.$$

III. Trigonometry

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos A \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$\sin A + \sin B = 2 \sin\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$$

$$\sin A - \sin B = 2 \sin\left(\frac{A-B}{2}\right) \cos\left(\frac{A+B}{2}\right)$$

$$\cos A - \cos B = -2 \sin\left(\frac{A+B}{2}\right) \sin\left(\frac{A-B}{2}\right)$$

$$\cos A + \cos B = 2 \cos\left(\frac{A+B}{2}\right) \cos\left(\frac{A-B}{2}\right)$$

$$\sin 2A = 2 \sin A \cos A.$$

$$\begin{aligned} \cos 2A &= 2 \cos^2 A - 1 = 1 - 2 \sin^2 A \\ &= \cos^2 A - \sin^2 A. \end{aligned}$$

$$\sin A/2 = \sqrt{(1 - \cos A)/2}$$

$$\cos A/2 = \sqrt{(1 + \cos A)/2}$$

$$\cos^2 A = (1 + \cos 2A)/2$$

$$\sin^2 A = (1 - \cos 2A)/2$$

$$\cos x = \frac{e^{jx} + e^{-jx}}{2}; \quad e^{jx} = \cos x + j \sin x$$

$$\sin x = \frac{e^{jx} - e^{-jx}}{2j}; \quad e^{jx} = \cos x + j \sin x$$

$$\sin(\omega t + \phi) = \cos(\omega t + \phi - 90^\circ)$$

$$\cos(\omega t + \phi) = \sin(\omega t + \phi + 90^\circ)$$

$$x \cos(\omega t + \phi_1) + y \cos(\omega t + \phi_2) = z \cos(\omega t + \phi_3)$$

Where $z = \sqrt{x^2 + y^2 + 2xy \cos(\phi_2 - \phi_1)};$

$$\phi_3 = \tan^{-1} \left[\frac{x \sin \phi_1 + y \sin \phi_2}{x \cos \phi_1 + y \cos \phi_2} \right]$$

Differentiation

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(\log_e x) = \frac{1}{x}$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\sinh x) = \cosh x$$

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$\frac{d}{dx}(ax+b)^n = n(ax+b)^{n-1} \cdot a$$

$$\frac{d}{dx} (a^x) = a^x \log_e a$$

$$\frac{d}{dx} (\cos x) = -\sin x$$

$$\frac{d}{dx} (\cot x) = -\operatorname{cosec}^2 x$$

$$\frac{d}{dx} (\log_a x) = \frac{1}{x \log a}$$

$$\frac{d}{dx} (\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} (\cosh x) = \sinh x$$

$$\frac{d}{dx} (\cot^{-1} x) = \frac{-1}{1+x^2}$$

$$\frac{d}{dx} = \frac{du}{dy} \times \frac{dy}{dx} \text{ (It is known as chain rule)}$$

$$\frac{d}{dx} (uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$D^n \log (ax+b) = (-1)^{n-1} \cdot (n-1)! a^n / (ax+b)^n$$

$$D^n (e^{mx}) = m^n \cdot e^{mx}$$

$$D^n (ax+b)^n = m(m-1)(m-2)(m-3) \dots (m-n+1) (ax+b)^{m-n}$$

$$D^n \begin{bmatrix} \sin(ax+b) \\ \cos(ax+b) \end{bmatrix} = a^n \begin{bmatrix} \sin(ax+b+n\pi/2) \\ \cos(ax+b+n\pi/2) \end{bmatrix}$$

Integration

$$\int \frac{1}{x} dx = \log_e x$$

$$\int \sin x dx = -\cos x$$

$$\int a^x dx = a^x / \log_e a$$

$$\int \tan x dx = -\log \cos x$$

$$\int \sec x dx = \log (\sec x + \tan x)$$

$$\int \sec^2 x \, dx = \tan x$$

$$\int \cos x \, dx = \sin x$$

$$\int \cot x \, dx = \log \sin x$$

$$\int \frac{1}{a^2 + x^2} \, dx = \frac{1}{a} \tan^{-1} \frac{x}{a}$$

$$\int \frac{1}{a^2 - x^2} \, dx = \frac{1}{2a} \log \frac{a+x}{a-x}$$

$$\int \frac{1}{x^2 - a^2} \, dx = \frac{1}{2a} \log \frac{x-a}{x+a}$$

$$\int \operatorname{cosec} x \, dx = \log (\operatorname{cosec} x - \cot x)$$

$$\int \operatorname{cosec}^2 x \, dx = -\cot x$$

$$\int (a^2 - x^2) \, dx = \frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$$

$$\int \sqrt{x^2 - a^2} \, dx = \frac{x\sqrt{x^2 - a^2}}{2} - \frac{a^2}{2} \cosh^{-1} \frac{x}{a}$$

$$\int \sqrt{a^2 + x^2} \, dx = \frac{x\sqrt{a^2 + x^2}}{2} + \frac{a^2}{2} \sinh^{-1} \frac{x}{a}$$

$$\int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx)$$

$$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$$

$$\int \sinh x \, dx = \cosh x$$

$$\int \cosh x \, dx = \sinh x$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} \, dx = \sin^{-1} \frac{x}{a}$$

$$\int \frac{1}{\sqrt{a^2 + x^2}} \, dx = \sinh^{-1} \frac{x}{a}$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} \, dx = \cosh^{-1} \frac{x}{a}$$

$$\int_0^{\pi/2} \sin^m x \cos^n x \, dx = \frac{(m-1)(m-3)(m-5) \dots \times (n-1)(n-3)(n-5) \dots}{(m+n)(m+n-2)(m+n-4) \dots} \times \frac{\pi}{2}$$

(if and only if m, n are even)

$$\int_0^{\pi/2} \sin^n x \, dx = \int_0^{\pi/2} \cos^n x \, dx = \frac{(n-1)(n-3) \dots}{n(n-2)(n-4) \dots} \times \left(\frac{\pi}{2}, \text{ if and only if } n \text{ is even} \right)$$

$$\int e^{ax} \, dx = \frac{e^{ax}}{a}; \quad \int e^x \, dx = e^x$$

$$\int x e^{ax} \, dx = \frac{e^{ax}}{a^2} (ax - 1); \quad \int x^n \, dx = \frac{x^{n+1}}{n+1}, \quad n \neq -1$$

$$\int x^2 e^{ax} \, dx = \frac{e^{ax}}{a^3} (a^2 x^2 - 2ax + 2)$$

$$\int e^{ax} \sin(bx) \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin(bx) - b \cos(bx))$$

$$\int e^{ax} \cos(bx) \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos(bx) + b \sin(bx))$$

$$\int \cos^2 ax \, dx = \frac{x}{2} + \frac{\sin 2ax}{4a}$$

$$\int x \cos(ax) \, dx = \left(\frac{1}{a^2} \right) (\cos(ax) + ax \sin(ax))$$

$$\int x^2 \cos(ax) \, dx = \left(\frac{1}{a^3} \right) (2ax \cos ax - 2 \sin ax + a^2 x^2 \sin ax)$$

$$\int \sin(ax) \, dx = -\left(\frac{1}{a} \right) \cos ax$$

$$\int \cos(ax) \, dx = \left(\frac{1}{a} \right) \sin ax$$

$$\int \sin^2(ax) \, dx = \left(\frac{x}{2} \right) - \frac{\sin 2ax}{4a}$$

$$\int x \sin(ax) \, dx = \left(\frac{1}{a^2} \right) (\sin ax - ax \cos ax)$$

$$\int x^2 \sin(ax) \, dx = \left(\frac{1}{a^3} \right) (2ax \cos ax + 2 \sin ax - a^2 x^2 \cos ax)$$

$$\int \cos(ax) \cos(bx) dx = \frac{\sin(a-b)x}{2(a-b)} + \frac{\sin(a+b)x}{2(a+b)}; (a^2 \neq b^2)$$

$$\int \sin(ax) \sin(bx) dx = \frac{\sin(a-b)x}{2(a-b)} - \frac{\sin(a+b)x}{2(a+b)}; (a^2 \neq b^2)$$

$$\int \sin(ax) \cos(bx) dx = -\left[\frac{\cos(a-b)x}{2(a-b)} + \frac{\cos(a+b)x}{2(a+b)} \right]; (a^2 \neq b^2)$$

$$\int_0^{\infty} \frac{\sin ax}{x} dx = \frac{\pi}{2}, 0, -\frac{\pi}{2} \text{ for } a > 0, a < 0, a = 0.$$

$$\int_0^{\infty} \frac{\sin^2 x}{x} dx = \pi/2$$

$$\int_0^{\infty} \frac{\sin^2 x}{x^2} dx = \pi/2$$

$$\int_0^{\infty} \frac{\sin^2 ax}{x^2} dx = |a| \pi/2$$

$$\int_0^{\infty} x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$$

$$\int_0^{\infty} e^{-r^2 x^2} dx = \frac{\sqrt{\pi}}{2r}$$

$$\int_0^{\infty} x e^{-r^2 x^2} dx = \frac{1}{2r^2}$$

$$\int_0^{\infty} x^2 e^{-r^2 x^2} dx = \frac{\sqrt{\pi}}{4r^3}$$

$$\int_0^{\infty} x^n e^{-r^2 x^2} dx = \frac{\Gamma[(n+1)/2]}{2 r^{n+1}}$$

$$\Gamma(k) = (k-1)!; \text{ for integers } k \geq 1.$$

IV Some Fourier Transform Theorems

Operation Performed	Function	Fourier Transform
Differentiation	$\frac{d^n \omega(t)}{dt^n}$	$(j 2\pi f)^n W(f)$
Integration	$\int_{-\infty}^t h(\lambda) d\lambda$	$(j 2\pi f)^{-1} H(f) + \frac{1}{2} H(0) \delta f$
Convolution	$a_1(t) * a_2(t) = \int_{-\infty}^{\infty} a_1(\lambda) \cdot a_2(t-\lambda) d\lambda$	$A_1(f) \cdot A_2(f)$
Multiplication	$a_1(t) a_2(t)$	$A_1(f) * A_2(f) = \int_{-\infty}^{\infty} A_1(\lambda) \cdot A_2(f-\lambda) d\lambda$
Linearity	$b_1 \omega_1(t) + b_2 \omega_2(t)$	$b_1 W_1(f) + b_2 W_2(f)$
Time delay	$\omega(t - T_d)$	$W(f) \cdot e^{-j 2\pi f T_d}$
Scale change	$\omega(at)$	$\frac{1}{ a } W\left(\frac{f}{a}\right)$
Conjugation	$\omega^*(t)$	$W^*(-f)$
Duality	$W(t)$	$\omega(+f)$
Translation for Real signal frequency	$\omega(t) \cos(\omega_c t + \theta)$	$\frac{1}{2} [e^{j\theta} W(f - f_c) + e^{-j\theta} W(f + f_c)]$
Translation for complex signal frequency	$\omega(t) e^{j\omega_c t}$	$W(f - f_c)$
Bandpass signal	$\text{Re} \{ P(t) e^{j\omega_c t} \}$	$\frac{1}{2} [P(f - f_c) + P^*(-f - f_c)]$

Some Fourier Transform Pairs

Function	Time waveform $\omega(t)$	Spectrum $W(f)$
Triangular	$\wedge\left(\frac{t}{T}\right)$	$T [\text{Sa}(\pi f T)]^2$
Rectangular	$\text{Rectan}\left(\frac{t}{T}\right)$	$T [\text{Sa}(\pi f T)]$
Unit step	$u(t) \triangleq \begin{cases} +1, & \text{for } t > 0 \\ -1, & \text{for } t < 0 \end{cases}$	$\frac{1}{2} \delta(t) + \frac{1}{j 2\pi f}$
Signum	$\text{Sig}(t) \triangleq \begin{cases} +1, & \text{for } t > 0 \\ -1, & \text{for } t < 0 \end{cases}$	$\frac{1}{j\pi f}$

Scalar constant	1	$\delta(f)$
$\sin c$	$\text{Sa}(2\pi W t)$	$\frac{1}{2W} \text{rectan}\left(\frac{f}{2W}\right)$
Impulse at time $t = t_0$	$\delta(t - t_0)$	$e^{-j2\pi f t_0}$
Sinusoid	$\cos(\omega_c t + \phi)$	$\frac{1}{2} e^{j\phi} \delta(f - f_c) + \frac{1}{2} e^{-j\phi} \delta(f + f_c)$
Gaussian	$e^{-\pi(t/t_0)^2}$	$t_0 e^{-\pi(f t_0)^2}$
One sided exponential	$\begin{cases} e^{-t/T}, & \text{for } t \geq 0 \\ 0, & \text{for } t < 0 \end{cases}$	$\frac{T}{1 + j2\pi f T}$
Two sided exponential	$e^{- t /T}$	$\frac{2T}{1 + (2\pi f T)^2}$
Impulse Train	$\sum_{k=-\infty}^{\infty} \delta(t - kT)$	$f_0 \sum_{m=-\infty}^{\infty} (\delta - mf_0), \text{ for } f_0 = \frac{1}{T}$

V. Some z-transform Pairs

Sequence	z-Transform	Region of convergence (ROC)
$\mu(n)$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
$\delta(n)$	1	For all values of z.
$\alpha^n \mu[n]$	$\frac{1}{1 - \alpha z^{-1}}$	$ z > \alpha $
$n \alpha^n \mu[n]$	$\frac{\alpha z^{-1}}{(1 - \alpha z^{-1})^2}$	$ z > \alpha $
$(n+1) \alpha^n \mu(n)$	$\frac{1}{(1 - \alpha z^{-1})^2}$	$ z > \alpha $
$(a^n \cos \omega_0 n) \mu(n)$	$\frac{1 - (a \cos \omega_0) z^{-1}}{1 - (2a \cos \omega_0) z^{-1} + a^2 z^{-2}}$	$ z > a $
$(a^n \sin \omega_0 n) \mu(n)$	$\frac{(a \sin \omega_0) z^{-1}}{1 - (2a \cos \omega_0) z^{-1} + a^2 z^{-2}}$	$ z > a $

VI. Some Important z-transform Properties

Property	Sequence	z-transform
Time Reversal	$m(-n)$	$M\left(\frac{1}{z}\right)$
Conjugation	$m^*(n)$	$M^*(z^*)$
Linearity ($m(n)$ and $h(n)$ are 2 sequences)	$\alpha m(n) + \beta h(n)$	$\alpha M(z) + \beta H(z)$
Time shifting	$m(n - n_0)$	$z^{-n_0} \cdot M(z)$
Convolution *	$m(n) * h(n)$	$M(z) H(z)$
Modulation	$m(n) h(n)$	$\frac{1}{2\pi j} \int_C M(v) H\left(\frac{z}{v}\right) v^{-1} dv$
Multiplication by exponention sequence	$\alpha^n g(n)$	$M\left(\frac{z}{\alpha}\right)$
Differentiation of $M(z)$	$n m(n)$	$-z \frac{dM(z)}{dz}$
Note :	$m(n)$ $h(n)$	$M(z)$ $H(z)$

VII. Some Standard Results**1) Permutaions and combinations :**

$${}^n P_r = \frac{n!}{(n-r)!}; \quad {}^n C_r = \frac{n!}{r!(n-r)!} = \frac{{}^n P_r}{r!}$$

$${}^n C_{n-r} = {}^n C_r; \quad {}^n C_0 = 1 = {}^n C_n$$

2) Progressions :

i) The numbers $b, br, br^2, br^3, \dots$ is said to be in a Geometric progression (G.P.);

It has n^{th} term $T_n = br^{n-1}$ and sum $s_n = \frac{b(1-r^n)}{1-r}$, $s_\infty = \frac{b}{1-r}$ ($r < 1$)

ii) The numbers $b, b+c, b+2c, \dots$ is said to be in Arithmetic progression (A.P.).

It has n^{th} term $T_n = b + (n-1)C$ and sum $S_n = \frac{n}{2} [2b + (n-1)c]$

iii) The numbers $b_1, b_2, b_3, \dots$ are said to be in harmonic progression (H.P.) if the,

$\frac{1}{b_1}, \frac{1}{b_2}, \frac{1}{b_3}, \frac{1}{b_4}, \dots$ are in Arithmetic progression.

iv) For two numbers a and b, their

- Arithmetic mean = $\frac{1}{2} (a + b)$
- Geometric mean = $\sqrt{ab}$
- Harmonic mean = $2ab/(a + b)$.

v) The Stirling's approximations :

When the 'n' is large then $n! \sim \sqrt{2\pi n} \cdot n^n \cdot e^{-n}$

vi) The first 'n' natural numbers are 1, 2, 3, 4, n.

Then

- $\sum n = \frac{n(n+1)}{2}$
- $\sum n^2 = \frac{n(n+1)(2n+1)}{6}$
- $\sum n^3 = \left\{ \frac{n(n+1)}{2} \right\}^2$

3) Important series :

i) Exponential series :

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \infty$$

ii) Log series :

$$\log (1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \infty$$

$$\log (1-x) = - \left[x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots \infty \right]$$

iii) sinx, cosx series :

a) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \infty$

b) $\sin hx = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \infty$

c) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \infty$

d) $\cos hx = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \infty$

- $\cos^2 A + \sin^2 A = 1$
- $\sin 2a = 2 \sin A \cos B$

$$= \frac{2 \tan A}{1 + \tan^2 A}$$
- $\cos 2a = \cos^2 A - \sin^2 A$

$$= 1 - 2 \sin^2 A$$

$$= 2 \cos^2 A - 1$$

$$= \frac{1 - \tan^2 A}{1 + \tan^2 A}$$
- $\tan (A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$
- $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
- $c \sin x + d \cos x = r \sin (x + \theta)$
 $c \cos x + d \sin x = r \sin (x - \theta)$

where

$$c = r \cos \theta$$

$$d = r \sin \theta, \text{ such that,}$$

$$r = \sqrt{c^2 + d^2}$$

$$\theta = \tan^{-1} \left(\frac{d}{c} \right)$$

- In any triangle ΔABC ,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

and

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

4) Binomial series :

$$\text{a) } (1+x)^n = 1 + nx + \frac{n(n-1)}{1 \cdot 2} x^2 + \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} x^3 + \frac{n(n-1)(n-2)(n-3)}{1 \cdot 2 \cdot 3 \cdot 4} x^4 + \dots \infty$$

$$\text{b) } (1+x)^{-n} = 1 - nx + \frac{n(n+1)}{1 \cdot 2} x^2 - \frac{n(n+1)(n+2)}{1 \cdot 2 \cdot 3} x^3 + \frac{n(n+1)(n+2)(n+3)}{1 \cdot 2 \cdot 3 \cdot 4} x^4 + \dots \infty$$

5) Tan x series :

$$\text{a) } \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \infty$$

$$\begin{aligned} \text{b) } \tanh^{-1} x &= \frac{1}{2} \log \frac{1+x}{1-x} \\ &= x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + \dots \infty \end{aligned}$$

Vectors

$$\text{i) } \mathbf{A} \cdot \mathbf{B} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\text{ii) } \mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{I} & \mathbf{J} & \mathbf{K} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}; \text{ where } \mathbf{A} = a_1 \mathbf{I} + a_2 \mathbf{J} + a_3 \mathbf{K} \text{ and } \mathbf{B} = b_1 \mathbf{I} + b_2 \mathbf{J} + b_3 \mathbf{K}$$

$$\text{iii) } \text{grad } f = \nabla f = \frac{\partial f}{\partial x} \mathbf{I} + \frac{\partial f}{\partial y} \mathbf{J} + \frac{\partial f}{\partial z} \mathbf{K}$$

$$\begin{aligned} \text{iv) } \text{div } \mathbf{F} &= \nabla \cdot \mathbf{F} \\ &= \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \end{aligned}$$

$$\begin{aligned} \text{v) } \text{curl } \mathbf{F} &= \nabla \times \mathbf{F} \\ &= \begin{vmatrix} \mathbf{I} & \mathbf{J} & \mathbf{K} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}; \text{ where } \mathbf{F} = f_1 \mathbf{I} + f_2 \mathbf{J} + f_3 \mathbf{K} \end{aligned}$$

Co-ordinate systems

Method	Cylindrical co-ordinates (ρ, ϕ, z)	Spherical co-ordinates (r, θ, ϕ)	Polar co-ordinates (r, θ)
Co-ordinate transformations	$x = \rho \cos \phi$ $y = \rho \sin \phi$ $z = z$	$x = r \sin \theta \cos \phi$ $y = r \sin \theta \sin \phi$ $z = r \cos \theta$	$x = r \cos \theta$ $y = r \sin \theta$
Jacobian	$\frac{\partial (x, y, z)}{\partial (\rho, \phi, z)} = \rho$	$\frac{\partial (x, y, z)}{\partial (r, \theta, \phi)} = r^2 \sin \theta$	$\frac{\partial (x, y)}{\partial (r, \theta)} = r$
Volume-element	$dV = \rho \, d\rho \cdot d\phi \, dz$	$dV = r^2 \sin \theta \, dr \, d\theta \, d\phi$	
Area elements	$dS_\rho = \rho \, d\phi \, dz$ $dS_\phi = dz \, d\rho$ $dS_z = \rho \, d\rho \cdot d\phi$	$dS_r = r^2 \sin \theta \, d\theta \, d\phi$ $dS_\theta = r \sin \theta \, d\phi \, dr \, ds_\phi$ $dS_\phi = r \, dr \cdot d\theta$	$dx \, dy = r \, d\theta \, dr$
Arc-element	$(dS)^2 = (d\rho)^2 + \rho^2 (d\phi)^2 + (dz)^2$	$(dS)^2 = (dr)^2 + r^2 (d\theta)^2 + (r \sin \theta)^2 (d\phi)^2$	$(ds)^2 = (dr)^2 + r^2 (d\theta)^2$

VIII. Some Basic Information For Engineering

1. Different systems of units.

Quantity	F. P. S. System	M. K. S. System	C. G. S. System
Mass	Pound (lb)	Kilogram (kg)	gram (gm)
Length	Foot (ft)	Metre (m)	Centimeter (cm)
Time	Second (sec)	Second (sec)	Second (sec)
Force	Lb. wt	Newton (nt)	Dyne

The MKS system mentioned above is also called as International system of Units (SI system).

2. Some important data

$\pi = 3.1416$	$1/\pi = 0.3183$	$\sqrt{\pi} = 1.7724$
$e = 2.7183$	$1/e = 0.3679$	$\sqrt{e} = 1.6487$
$\sqrt{2} = 1.4142$	$\sqrt{3} = 1.732$	$\sqrt{10} = 3.1623$
$\log_e^2 = 0.6931$	$\log_e^3 = 1.0986$	$\log_e^{\pi} = 1.1447$
$\log_e^{10} = 2.3046$	$\log_{10}^e = 0.4343$	
$1 \text{ Rad} = 57^\circ 17' 45''$	$1^\circ = 0.0174 \text{ Rad.}$	

3. Conversion factors.

1.	1 Nautical mile = 6080.2 ft = 1.853 km.
2.	1 ft = 30.48 cm = 0.3048 m
3.	1 m = 100 cm = 3.2804 ft
4.	$1 \text{ ft}^2 = 0.0929 \text{ m}^2$
5.	$1 \text{ ft}^3 = 0.0283 \text{ m}^3$
6.	1 acre = 4840 yd ² = 4046.77 m ²
7.	$1 \text{ m}^3 = 35.32 \text{ ft}^3$
8.	1m/sec = 3.2804 ft/sec
9.	1 mile/h = 1.609 km/h = 0.869 knots. (Nautical miles/hr)
10.	1 Calorie (Cal) = 4.1840 joules 1 joule = 10 ⁷ ergs
11.	1 horse power (hp) = 178.2 Cal/sec = 0.7457 kw
12.	1 kW = 1000 watts (W) = 238.9 cal/sec
13.	1gravity = 32 ft/sec ² = 980 cm/sec ²
14.	$^{\circ} \text{F} = (^{\circ} \text{C} \times 1.8) + 32$

4. Important prefixes used.

Some multiples and submultiples	Symbols used	Prefixes
10 ¹²	T	tera
10 ⁹	G	giga
10 ⁶	M	Mega
10 ³	K	Kilo
10 ²	h	hecto
10	da	deca
10 ⁻¹	d	deci
10 ⁻²	c	Centi
10 ⁻³	m	milli
10 ⁻⁶	μ	micro
10 ⁻⁹	n	nano
10 ⁻¹²	p	pico

Note : The prefix terms deci and centi are used with metre (Centimetre is a standardized unit of length).

5. Greek letters.

α	alpha
β	beta
γ	gamma
δ	delta
ϵ	epsilon
ι	iota
θ	theta
ϕ	phi
ψ	psi
η	eta
ξ	xi
ζ	zeta
λ	lambda
μ	mu
π	pi
ρ	rho
σ	sigma
τ	tau
ω	omega
χ	chi
Γ	Capital gamma
Δ	Capital delta
Σ	Capital sigma

6. Useful constants**i) Physical constants :**

- 1) Plank's constant $h = 6.626 \times 10^{-34}$ joule-sec.
- 2) Boltzmann's constant $k = 1.38 \times 10^{-23}$ joule/deg.kelvin.
- 3) Electron charge $q = 1.602 \times 10^{-19}$ coulomb.
- 4) Speed of light in vacuum $c = 2.998 \times 10^8$ m/sec.
- 5) Standard (absolute) temp $T_0 = 273$ deg kelvin.
- 6) Thermal voltage $v_T = 0.026$ volt at room temp.
- 7) One hertz (hz) = 1 cycle/sec
where 1 cycle = 2π radians.
- 8) One watt (W) = 1 joule/sec.

ii) Mathematical constants :

- 1) Base of natural logarithm $e = 2.718281$
- 2) Logarithm of 2 to base e $\log_e 2 = 0.69314$
- 3) Logarithm of e to base 2 $\log_2 e = 1.4426$
- 4) Logarithm of 2 to base 10 $\log_{10} 2 = 0.30103$
- 5) Pi or $\pi = 3.141592$

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